Characteristics of polynomial functions

1 Given $P(x) = 4x^5 + 3x^6 - 8$, find the following:

a $P(0)$ **b** $P(1)$ **c** $P(-4)$ **d** $P(4)$

2 Given $P(x) = 4 - \dfrac{7x^6}{6}$, find the following:

a $P(-1)$ **b** $P\left(\sqrt{2}\right)$ **c** $P\left(\dfrac{1}{2}\right)$ **d** $P(0)$

3 Given $P(x) = \left(x^4 + 5\right)\left(4 - 3x^5\right)$, find the following:

a $P(0)$ **b** $P\left(-\dfrac{1}{3}\right)$ **c** $P(2)$ **d** $P(-1)$

4 Consider $f(x) = -x^2$.

a Does the graph rise or fall to the right? **b** Does the graph rise or fall to the left?

5 Patricia claims she can graph a third-degree polynomial function with 3 turning points. Is this possible or impossible? Explain.

6 Determine the relationship between the degree of a polynomial function and the number of turning points on its graph.

7 Consider $f(x) = -6x^2 - 4x + 5$ which is concave down.

a State the coordinates of the y-intercept of the function.

b How many x-intercepts does the function have?

8 Is it possible for the graph of a polynomial function to have:

a No x-intercepts? **b** No y-intercepts?

9 Determine whether the following statement is true or false:
"Even-degree polynomial functions have graphs with opposite behavior at each end."

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10 Given that $f(x)$ is an even function, state whether the following are also even functions:

a $f(x+3)$ **b** $f(x+4)+1$ **c** $f(x-2)$ **d** $f(x)+1$

Graphs of polynomial functions

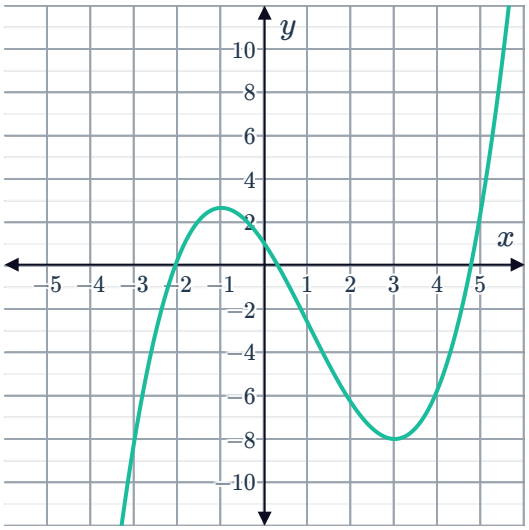
11 For each of the following functions:

i State the x-values of any relative maxima or minima of $f(x)$.

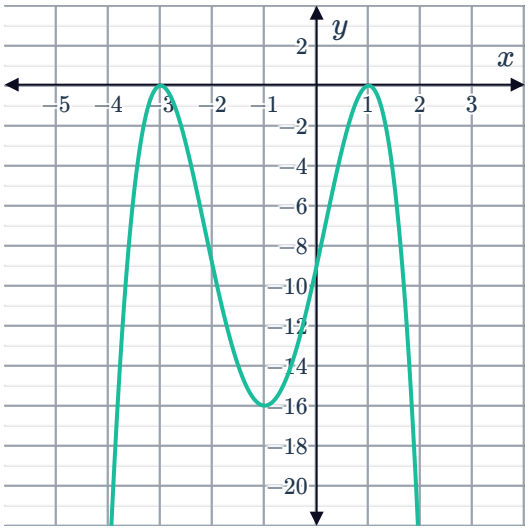
ii Find the regions of the domain where $f(x)$ is increasing.

iii Find the regions of the domain where $f(x)$ is decreasing.

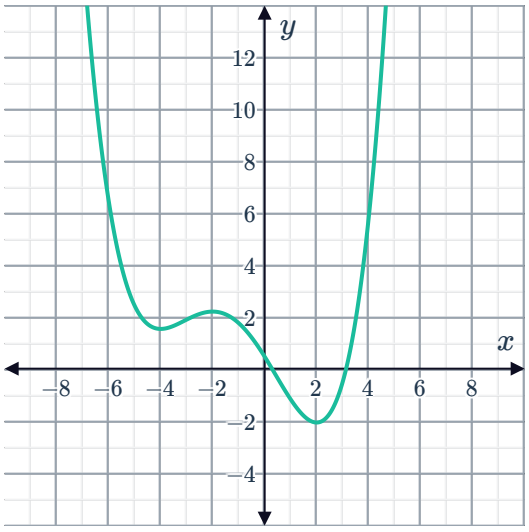
a



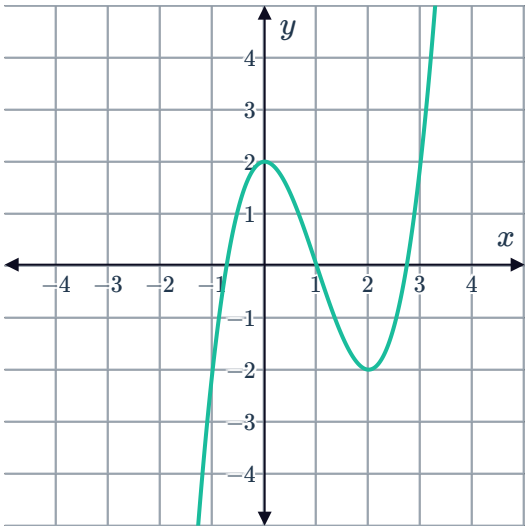
b



c



d



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12 Consider the function which has intercepts at $(-4, 0)$, $(2, 0)$ and $(0, 3)$.

a State the least degree of a polynomial that goes through these points.

b Sketch the graph of the function.

13 For each of the following functions:

i Find the x-intercepts.

ii Find the y-intercept.

iii Sketch the graph of the function.

a $y = (x+3)(x+2)(x-2)$

b $y = -(x+4)(x+2)(x-1)$

c $y = (x-2)^2(x+5)$

d $y = -(x-1)^2(x+2)$

e $y = (x-1)(x-2)(x+4)(x+5)$

f $y = -(x+1)(x+3)(x+4)(x-4)$

g $y = (x-2)^2(x+3)(x-1)$

h $y = -(x-3)^2(x+1)(x-2)$

i $y = (x-1)^2(x-4)^2$

j $y = -(x-3)^2(x+2)^2$

14 Consider $y = x^3 - 6x^2 + 3x + 10$.

a Does the graph rise or fall to the right?

b Does the graph rise or fall to the left?

c Express the equation in factored form.

d Find the x-intercepts.

e Find the y-intercept.

f Sketch the graph of the function.

15 Consider $y = x^4 - 4x^2$.

a State the leading coefficient.

b Does the function rise or fall to the left?

c Does the function rise or fall to the right?

d Express the equation in factored form.

e Find the x-intercepts.

f Find the y-intercept.

g Sketch the graph of the function.

16 Consider the polynomial function $y = x^4 - x^2$.

a State the leading coefficient.

b Is the degree odd or even?

c Does the function rise or fall to the left?

d Does the function rise or fall to the right?

e Sketch the graph of the function.

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17 For each of the following functions:

i As $x \to -\infty$, what does y approach?

ii As $x \to \infty$, what does y approach?

iii Find the x-intercepts.

iv Find the y-intercept.

v Sketch the graph of the function.

a $y = x\left(x+1\right)\left(x-2\right)\left(x-4\right)$

b $y = -x\left(x+2\right)\left(x-2\right)\left(x-3\right)$

c $y = x^5 - x^2 + 2$

d $y = -x^5 + x^3$

18 Sketch the graph of the following functions:

a $f\left(x\right) = 2x^2 - x - 15$

b $f\left(x\right) = x\left(x+3\right)\left(x-3\right)$

c $f\left(x\right) = x^3 + 2x^2 - 4x - 8$

d $f\left(x\right) = -x^3 + x^2 - 3x + 4$

e $f\left(x\right) = 2x^4 - x^2 + 2$

f $f\left(x\right) = x^5 + \frac{1}{10}x - 3$

g $f\left(x\right) = x^5 - x^3$

h $f\left(x\right) = -x^6 + 2x^5 - 6x^2$

Applications

19 The recommended level of good cholesterol in women, in $\mathrm{mg/dL}$, can be modeled by the function $f\left(x\right) = 0.13x + 51.25$, where x is measured in years.

a Find and interpret $f\left(23\right)$.

b Find and interpret $f\left(51\right)$.

20 Eileen planted a mango tree in 1970. The number of liters of fertilizer her mango tree needs each year can be modeled by the function $F\left(x\right) = 0.303x + 12.113$, where x is the number of years after the tree is planted.

a Find the amount of fertilizer the tree will need in 1985.

b Find the amount of fertilizer needed in 2005.

c How many full years after planting will the tree need at least $19.45\ \mathrm{L}$ of fertilizer?

d In what year will Eileen's mango tree need at least $19.45\ \mathrm{L}$ of fertilizer?

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21 The population, P, of the endangered American Burying Beetle in 1940 was approximately $12\,580$. In 1980, the population was approximately 1405. The population of the American Burying Beetle can be estimated using the function $P(x) = -269x + 14\,800$, where x is the number of years after 1930.

a Estimate the size of the beetle population in 1950.

b Estimate the size of the beetle population in 1981.

c How many whole years after 1930 will the American Burying Beetle become extinct according to the model?

d In what year would the American Burying Beetle become extinct according to the model?

22 The length of the growing season in different regions can be estimated by counting the average number of consecutive frost free days in that region. The elevation, E, in meters, above sea level at which there are, on average, d frost free days is given by the function:

$$E(d) = \frac{1}{2}(160 - d) + \frac{1}{10}(160 - d)^2$$

a At what elevation are there 100 frost free days on average?

b At what elevation are there 160 frost free days on average?

23 A safety flare is projected up into the air and its height $h(t)$ feet above the ground, t seconds after it is projected is given by $h(t) = 105t - 15t^2$.

a Determine the height of the flare 3 seconds after it is projected.

b How long will it take for the flare to reach the ground?

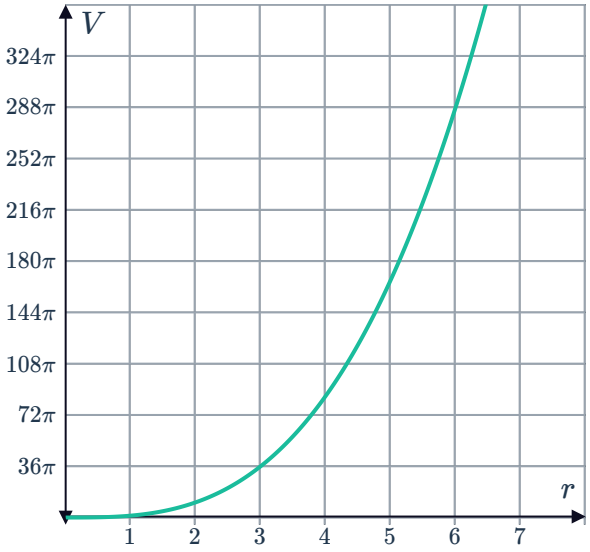
24 The annual cost for a product is given by $C = 7x + 2x^2$ and the annual revenue is given by $R = 7 - 3x + 4x^2 + x^3$, where x is the number of units produced and sold, and R and C are the revenue and cost respectively in thousands of dollars.

a Form an expression for P, the net profit after producing and selling x units.

b After selling 600 units, is the company making a profit or loss?

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25 The volume of a sphere has the formula $V = \frac{4}{3}\pi r^3$. The graph relating r and V is shown:



a Complete the following table of values for $V = \frac{4}{3}\pi r^3$, in terms of π:

r	1	2	4	5	7
V					

b State whether each of the following is true for the volume, V, of a sphere that has a radius measuring 4.5 m:

i $20\pi < V < 57\pi$

ii $85\pi < V < 166\pi$

iii $179\pi < V < 696\pi$

iv $221\pi < V < 366\pi$

c Using the graph, find the radius of a sphere of volume 288π m^3.

26 The mass, in kilograms, of a cubic container of water is given by $M = 0.001r^3$, where r is the side length of the cube shaped container in centimeters.

a Find the mass of the container of water if it has side lengths of 20 cm.

b The greatest mass Jimmy can carry is equivalent to his own weight, which is 88 kg. Find the length of the largest cubic container of water Jimmy can carry, correct to two decimal places.

c A fish tank requires several identical cubic containers to be delivered, each with side length 20 cm. Due to workplace safety requirements, workers cannot handle a total weight heavier than 172 kg. Find the most number of containers that can be delivered.

27 A box without a top cover is to be constructed from a rectangular cardboard, measuring 6 ft by 10 ft by cutting out four square corners of length x ft. Let V represent the volume of the box.

a Express the volume V of the box in terms of x, writing the equation in factored form.

b For what range of values of x is the volume function defined?

c Sketch the graph of the volume function.

d Determine the volume of a box that has a height equivalent to the shorter dimension of the base.

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